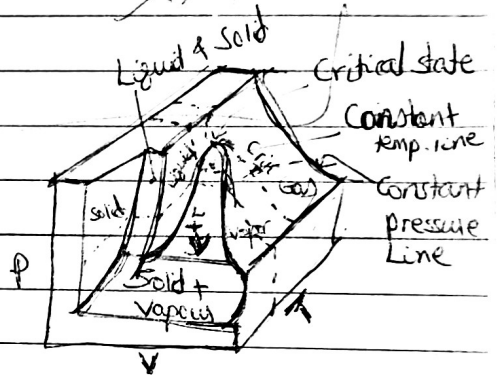
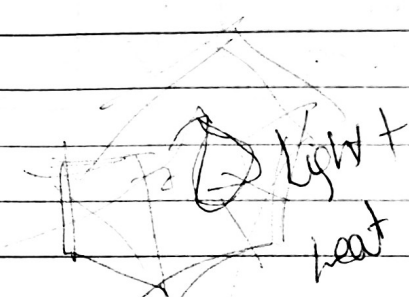
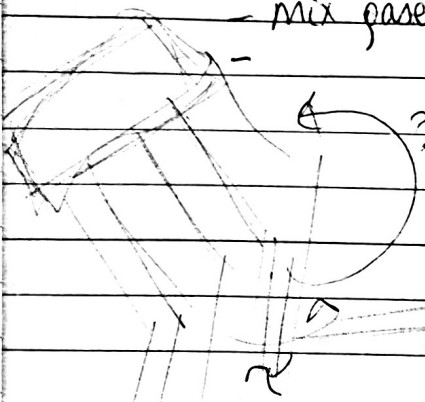


Thermodynamic Cycles

"The total entropy change (system + surroundings) is always greater than or equal to zero for any change of state of the system."

- Efficient processes possible for converting energy from one form to another are processes where $\Delta S = 0$
- Natural processes tend to go in certain directions than vice versa
- Direction → ability to do work with a system.
- mix gases than unmix



→ Reversible?

→ Irreversibility & inefficiency

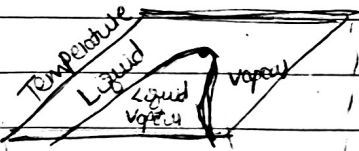
Thermodynamic cycles

Refrigeration

Power cycles

Idealised models of real cycles

Ideal cycles
• Carnot cycles



Gas power cycles

Vapour power cycles

Internal combustion Engines

External combustion Engines

→ External combustion Engines

• Otto cycle

• Ericsson cycle

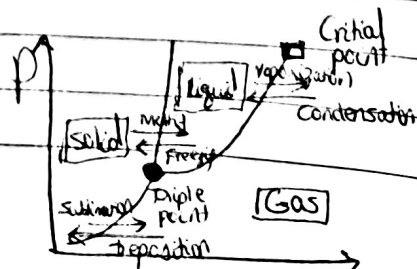
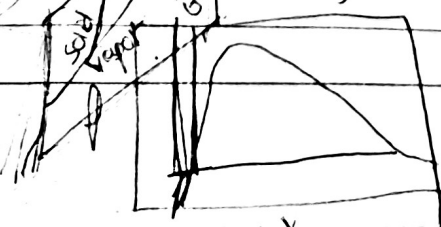
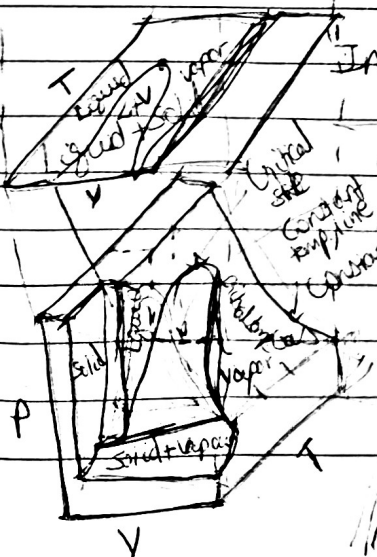
• Rankine cycle

• Diesel cycle

• Stirling cycle

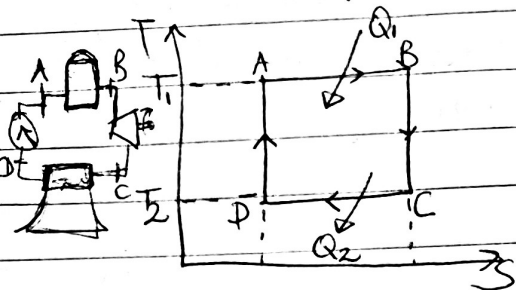
• Organic

• Gas turbine engines (Brayton cycle)



Carnot cycle

- Upper limit of both energetic & exergetic thermodynamic efficiency



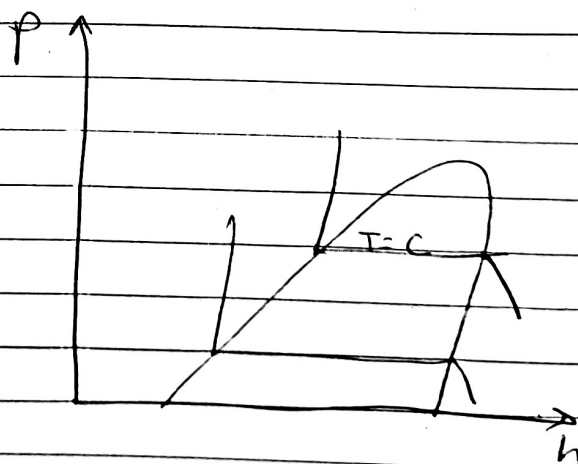
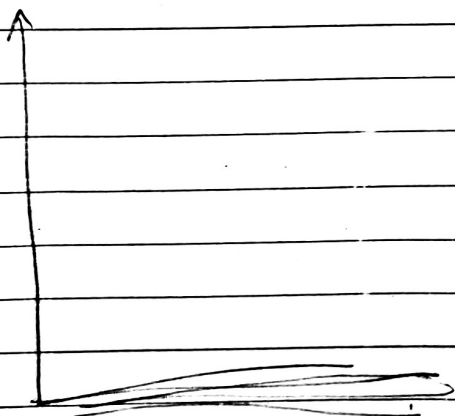
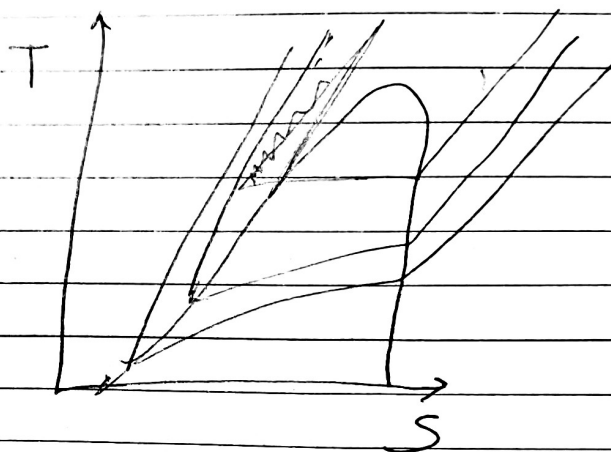
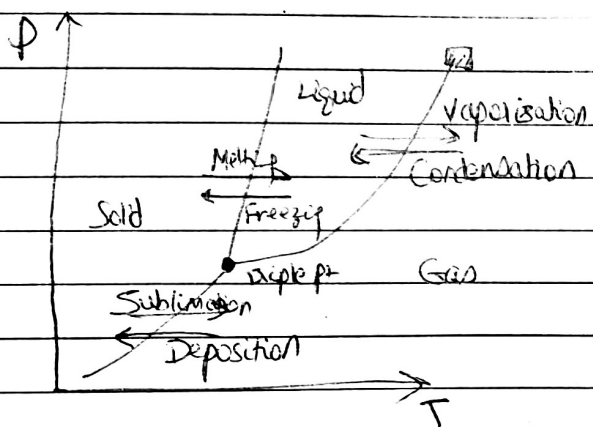
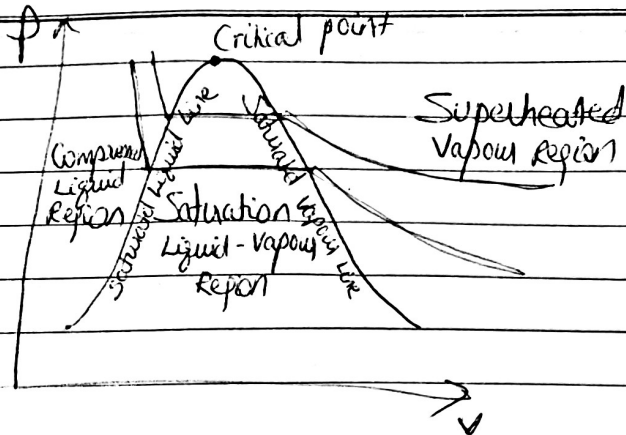
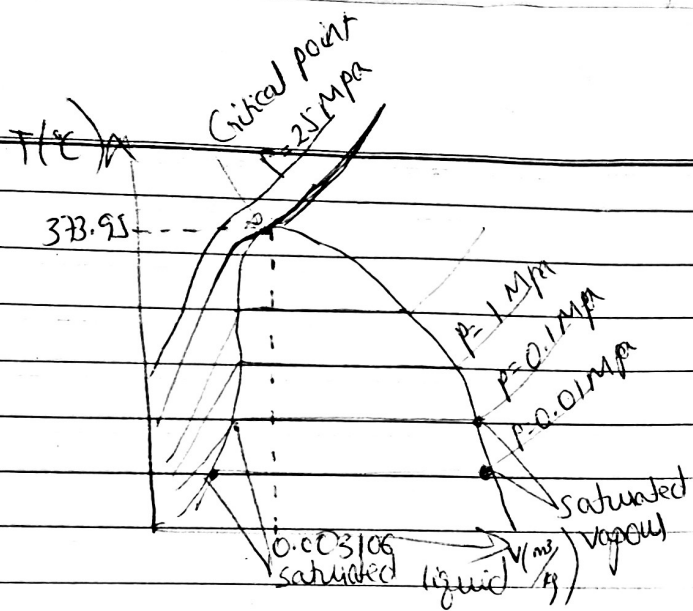
AB - Isothermal Expansion

BC - Isentropic exp expansion

CD - Isothermal compression

DA - Isentropic compression

$$\eta = 1 - \frac{T_2}{T_1}$$



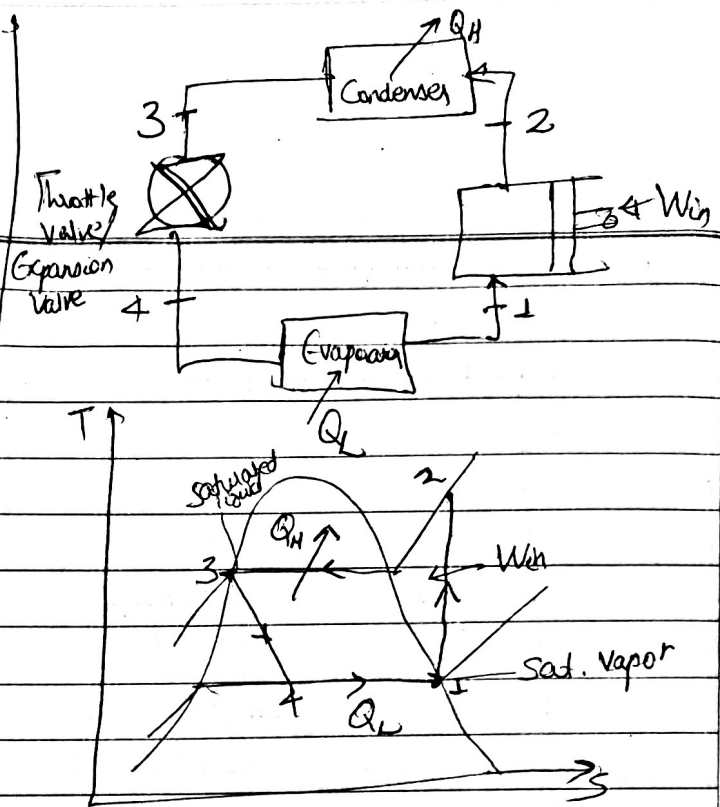
1. Vapor power cycle

Rankine cycle

4 - processes

→ Efficiency (η_m)

How to increase :- Reheat



2. Gas power cycle

→ Gasoline

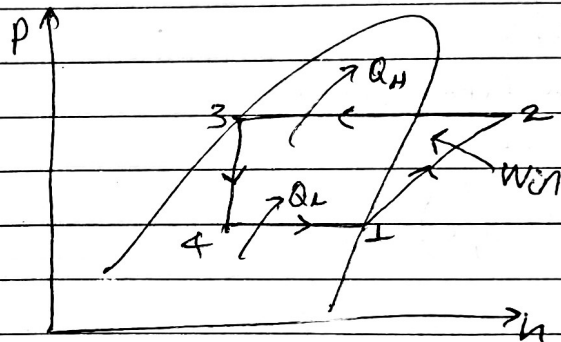
→ Diesel

→ Brayton

→ η

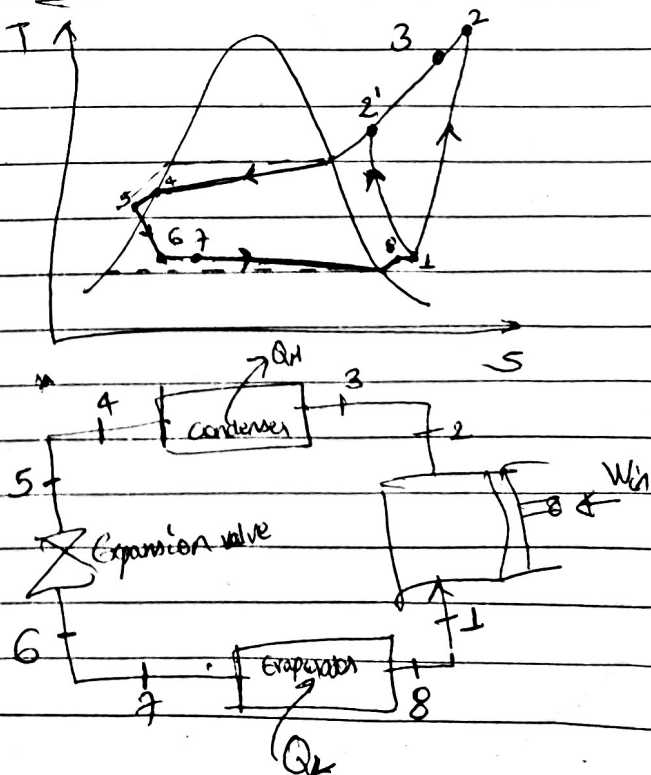
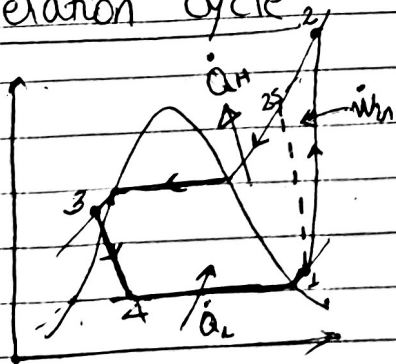
→ How to increase

→ Jet-propulsion cycles



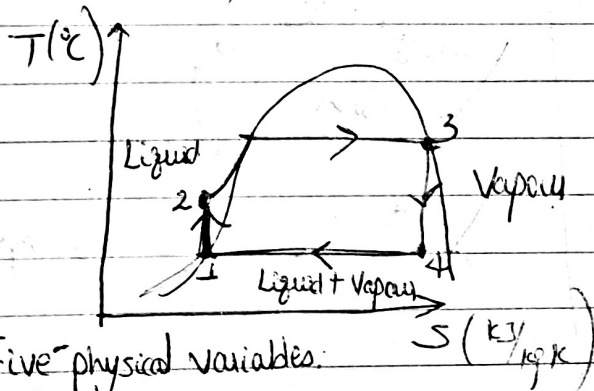
Actual

3. Refrigeration cycle



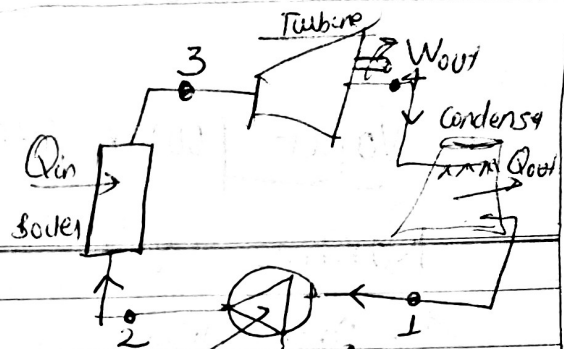
1. Vapour power cycle / Rankine cycle

Ideal cycle / Rankine



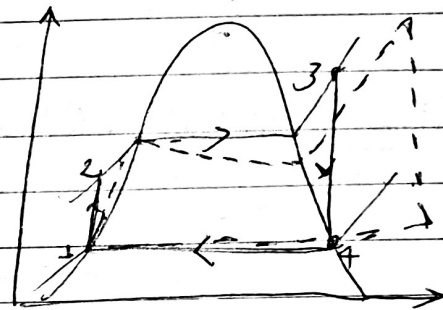
Five physical variables:

1. P (Pa)
2. T (K)
3. x (steam quality)
4. h (kJ/kg)
5. s (kJ/kg.K)

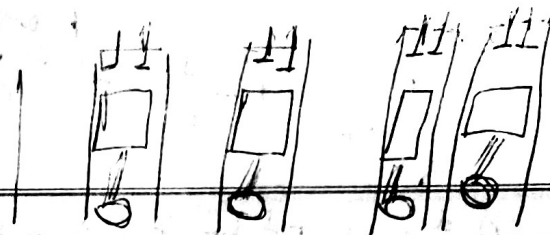


Actual Rankine cycle

- considering imperfections & irreversibility

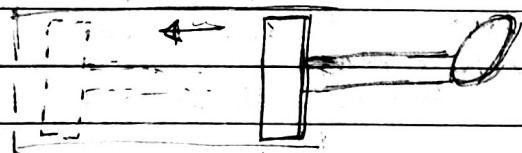


Gas power cycle



1. Otto cycle

- Our goal is not to get the actual engine thermal efficiency but trying to get the maximum achievable
- 100% $\eta \rightarrow$ violates 2nd law of TTD
- Cannot $\eta \rightarrow$ inachievable



Stroke
TDC BDC
 V_2 V_1

$r =$ compression ratio $= \frac{V_1}{V_2}$

① - Maximum pressure

- In compression: ideal is constant entropy / isentropic

② Heat addition for ignition

In real life - spark plug

In ideal cycle - $\square$ heat reservoir

Piston in contact to high heat reservoir

Heat addition in closed system $\rightarrow$ Increase internal energy / $\rightarrow$ Provide work

So neglecting one will maximize the other

$\rightarrow$ Expansion is Isochoric / constant volume

$\rightarrow$ Piston doesn't move / a little in real life

$\rightarrow$ Not to lose heat to work but to internal energy $\rightarrow T \uparrow$

3. Expansion



TDC BDC

- Work to get
- No friction, no energy loss $\rightarrow$ Isentropic

4. Heat rejection / 2nd law TTD

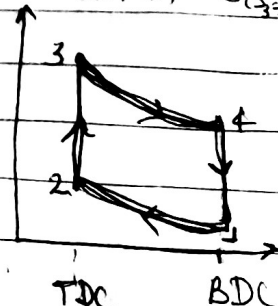
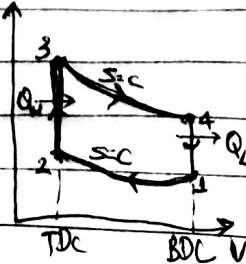
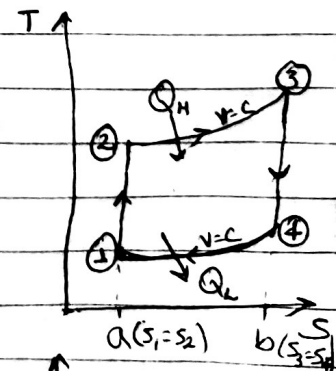
Low temp reservoir - heat

Ideal cycle

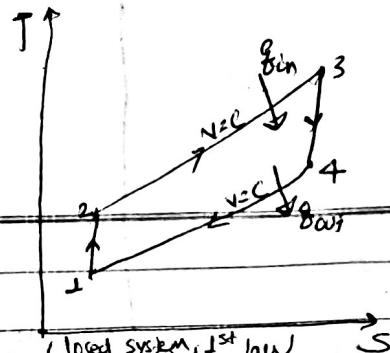
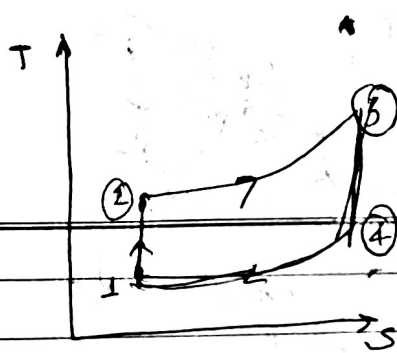
- Ideal processes:
 - Compression $\rightarrow$ BDC $\rightarrow$ V_1 (Stroke ideal is isentropic)
 - Heat Addition
 - Expansion
 - Heat Rejection

Assumptions

- Since do (air-fuel combustion) is complex
- Air is the working fluid & is an ideal gas
- Closed system: same mass
- Exhaust & air intake $\rightarrow$ heat transfer
- Combustion is - heat transfer
- All processes are internally reversible
- Gas specific heat is constant
- cycle does not involve any friction
- Pipes connecting components have no heat loss
- Kinetic & potential energy changes are neglected



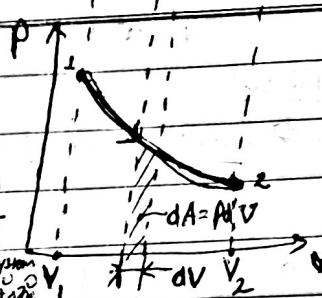
From Thermo I



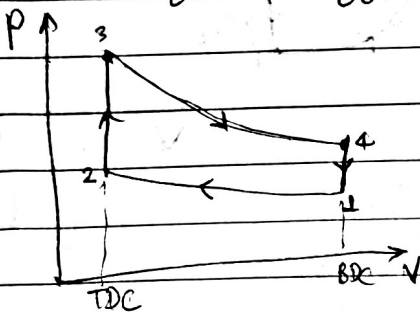
$$\eta = \frac{W_{net}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

Closed system, 1st law
 $W_{net} = Q_{net}$ For cycle
 $= Q_{in} - Q_{out}$

Closed system
 1st law
 $E_{in} - E_{out} = \Delta E_{system}$
 $Q - W = \Delta U + \Delta KE + \Delta PE$



Boundary work is path dependent

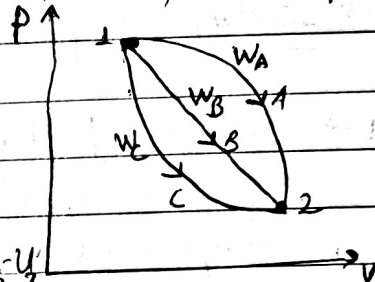


Process 1-2
 Isentropic compression
 $(Q_{in} + W_{in}) - (Q_{out} + W_{out}) = \Delta U = U_2 - U_1$

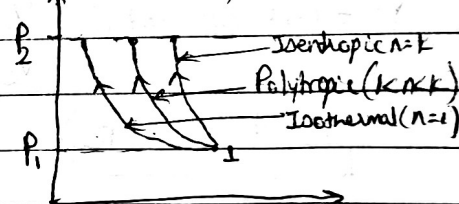
Process 2-3 ($V_2 = V_3$)
 Isochoric heat addition
 No work ($Q_{in} + W_{in}) - (Q_{out} + W_{out}) = U_3 - U_2 = C_v(T_3 - T_2)$

Process 3-4
 Isentropic expansion
 $(Q_{in} + W_{in}) - (Q_{out} + W_{out}) = U_4 - U_3$

Process 4-1 ($V_4 = V_1$)
 Isochoric heat rejection
 No work ($Q_{in} + W_{in}) - (Q_{out} + W_{out}) = U_1 - U_4 = C_v(T_4 - T_1)$



Three internally reversible cases



Process 4-1 ($V_4 = V_1$)
 Isochoric heat rejection
 No work ($Q_{in} + W_{in}) - (Q_{out} + W_{out}) = U_1 - U_4 = C_v(T_4 - T_1)$

During actual compression & expansion processes of gases: P & V related by $PV^n = C$; n is γ for isentropic, n is k for polytropic process

$$\eta = 1 - \frac{1}{r^{k-1}}$$

Efficiency calculation without engine stat

2 approaches
 ① Constant specific heats (Approximate Analysis) (Ideal gas)
 $TV^{k-1} = C$
 $TP^{(1-k)/k} = C$
 $PV^k = C$

② Variable sp. heats (Exact Analysis) (Ideal gas)
 Steady: when ΔT is small & C_p & C_v vary
 $S_2 = S_1 + R \ln \frac{P_2}{P_1}$
 $S_2 = S_1 + R \ln \frac{P_2}{P_1}$

Steady: when ΔT is small & C_p & C_v vary
 $S_2 = S_1 + R \ln \frac{P_2}{P_1}$
 $S_2 = S_1 + R \ln \frac{P_2}{P_1}$

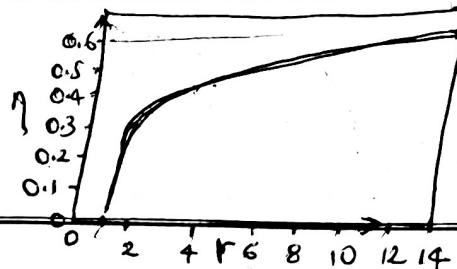
$\eta = 1 - \left(\frac{V_1}{V_2}\right)^{1-k}$
 $= 1 - \left(\frac{P_2}{P_1}\right)^{\frac{1-k}{k}}$

Further analyzing
 $\eta = 1 - \frac{T_1}{T_2} \left(\frac{T_4 - T_1}{T_3 - T_2}\right)$
 $= 1 - \frac{T_1}{T_2} = f(T_1, T_2)$

Select a good thermocouple & measure all.
 $V_1 = V_4$; $V_2 = V_3$
 $\frac{T_1}{T_2} = \frac{P_1}{P_2}$

$$\eta_{\text{otto}} = 1 - \frac{1}{r^{k-1}} = f(r, k)$$

compression ratio
Heat ratio



k=1.4
fixed

$\eta \uparrow$ as $r \uparrow$ but there is a limit

Otto ignition temp

$$r = \frac{V_1}{V_2} \Rightarrow r \uparrow \Rightarrow \frac{V_1 \uparrow}{V_2 \downarrow} \rightarrow \text{Big massively}$$

$V_2 \downarrow \rightarrow V_2$ is very small $\rightarrow$ Ignition temp rises dramatically

$\rightarrow$ May exceed auto ignition temp

$\rightarrow$ If the fuel burns by itself without no control

$\rightarrow$ No control on timings

$\rightarrow$ Exactly when to fire

Auto Ignition temp

a temp, at which
auto ignition occurs
without external
heat source

eg paper

216, 218

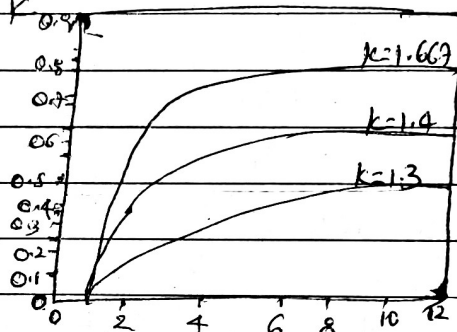
well burn

$\rightarrow$ Otto cycle ignition temp
of fuel

$\rightarrow \eta \uparrow$ by changing type of gas: k

Argon $k=1.667 \rightarrow \eta \uparrow$ than Air $k=1.4$

Availability, combustion is a question.



* knock sensor: detects pre-ignition
or knocking.

$\rightarrow$ combustion processes

Starts too early

$\rightarrow$ Pre-ignition is very rough for engine.

* 35% of cost of a new car goes to electronics.

Programmed chip

≡ Analog input sensors
≡ Analog outputs to
actuators

Temp
rpm

Oxygen

$\rightarrow$ Better processing;
better ICE.

2. Diesel cycle

Fuel - Coal dust
- Vegetable oil
- Petro diesel

Process

1-2 $\rightarrow$ Isentropic compression

2-3 $\rightarrow$ Isobaric heat addition

$$Q_{in} = C_p(T_3 - T_2)$$

$\uparrow$ constant p

3-4 $\rightarrow$ Isentropic expansion

4-1 $\rightarrow$ Isochoric heat rejection

$$Q_{out} = C_v(T_4 - T_1)$$

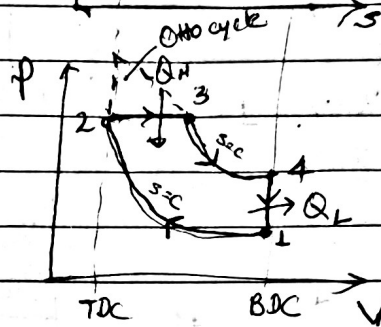
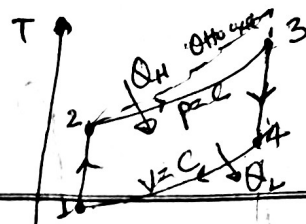
$$r = \text{compression ratio} = \frac{V_1}{V_2}$$

$$\text{but } V_1 = V_4; V_2 \neq V_3$$

$$\text{So } r_c = \text{cut-off ratio} = \frac{V_3}{V_2} \text{ characteristic}$$

$$\Rightarrow \boxed{\text{Diesel} = f(r, r_c)}$$

$$r_c \frac{T_1}{T_4} = r_c^{1-k} \text{ or } \frac{T_1}{T_4} = r_c^{-k} \& \boxed{\frac{T_4}{T_1} = r_c^k}$$



$$\Rightarrow \eta = 1 - \frac{Q_{out}}{Q_{in}} = \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_2)}$$

$$= 1 - \frac{C_v}{C_p} \frac{T_1}{T_2} \left(\frac{T_4}{T_1} - 1 \right)$$

$$= 1 - \frac{1}{k} \frac{T_1}{T_2} \left(\frac{T_4}{T_1} - 1 \right) \left(\frac{T_3}{T_2} - 1 \right) \left(\frac{T_4}{T_1} - 1 \right) \left(\frac{T_3}{T_2} - 1 \right)$$

need all measurements

Further analyzing

$$\eta_{\text{Diesel}} = 1 - \frac{1}{k} \frac{1}{r^{k-1}} \left(\frac{r_c^k - 1}{r_c - 1} \right)$$

$$= f(r, r_c)$$

Otto cycle

$$r = 10$$

$$r_c = X (\text{none})$$

$$\eta_{\text{Otto}} = 60.2\%$$

Limited by autoignition temp

Diesel cycle

$$r = 10$$

$$r_c = 2$$

$$\eta_{\text{Diesel}} = 53.7\%$$

$$r = 20$$

$$r_c = 2$$

$$\rightarrow \eta_{\text{Diesel}} = 64.7\%$$

- It can reach high compression ratio since air is compressed

- More efficient than Otto b/c r can be high (increased)

Typical SI engines have r b/w 6:1 & 12:1

For $k = 1.4$

$$6:1 (r=6) \quad \eta_{\text{Otto, th}} = 0.5116 (51.16\%)$$

$$12:1 (r=12) \quad \eta_{\text{Otto, th}} = 0.6299 (62.99\%)$$

In reality: $\eta_{\text{Otto}} = 20\% - 30\%$

$\uparrow r \rightarrow \eta_{\text{Otto}} \rightarrow$ r is limited by combustion & autoignition

